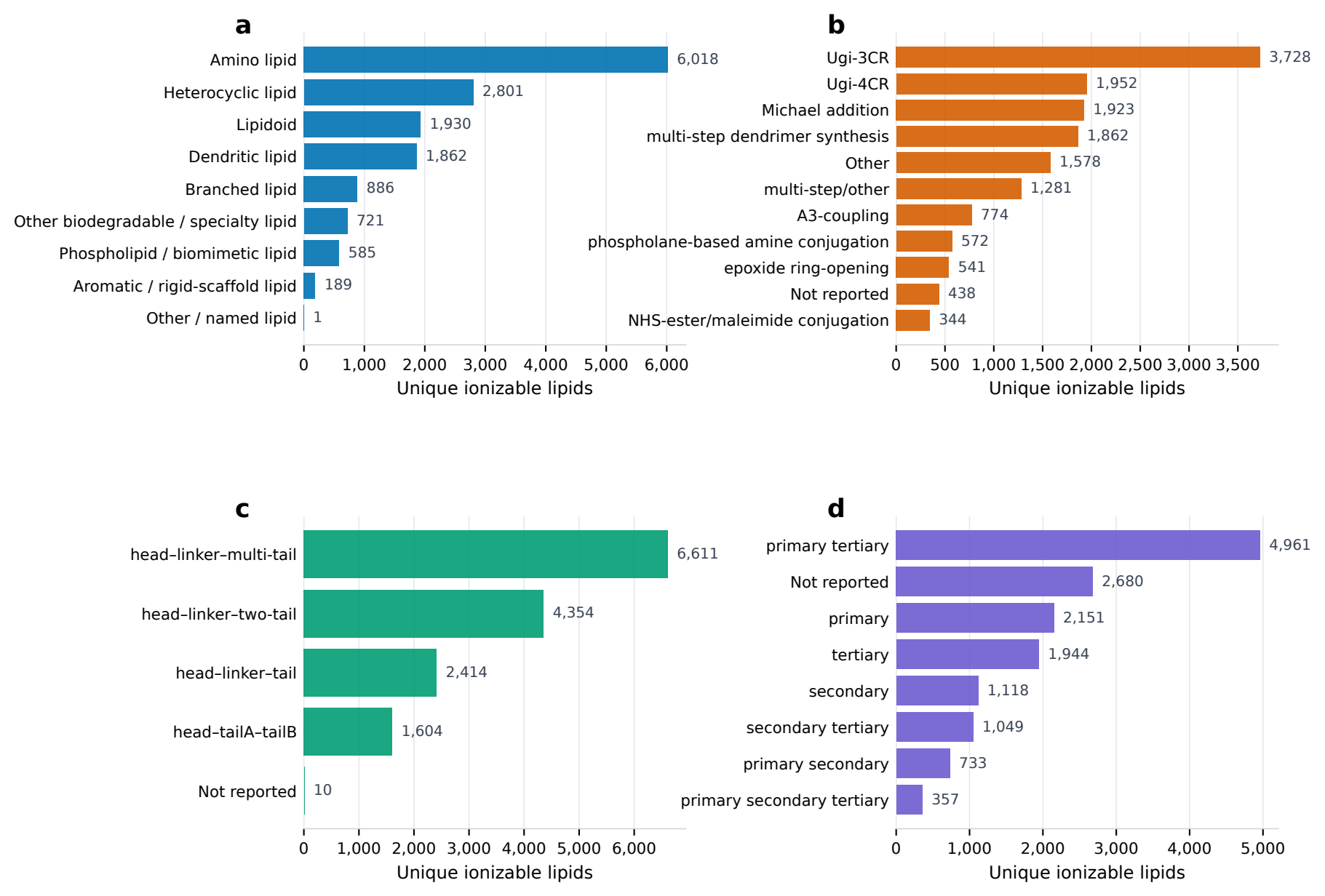
a
Amino lipid
6,018
Heterocyclic lipid
2,801
Lipidoid
1,930
Dendritic lipid
1,862
Branched lipid
886
Other biodegradable / specialty lipid
721
Phospholipid / biomimetic lipid
585
Aromatic / rigid-scaffold lipid
189
Other / named lipid
1
0
1,000
2,000
3,000
4,000
5,000
6,000
Unique ionizable lipids
b
Ugi-3CR
3,728
Ugi-4CR
1,952
Michael addition
1,923
multi-step dendrimer synthesis
1,862
Other
1,578
multi-step/other
1,281
A3-coupling
774
phospholane-based amine conjugation
572
epoxide ring-opening
541
Not reported
438
NHS-ester/maleimide conjugation
344
0
500
1,000
1,500
2,000
2,500
3,000
3,500
Unique ionizable lipids
c
head–linker–multi-tail
6,611
head–linker–two-tail
4,354
head–linker–tail
2,414
head–tailA–tailB
1,604
Not reported
10
0
1,000
2,000
3,000
4,000
5,000
6,000
Unique ionizable lipids
d
primary tertiary
4,961
Not reported
2,680
primary
2,151
tertiary
1,944
secondary
1,118
secondary tertiary
1,049
primary secondary
733
primary secondary tertiary
357
0
1,000
2,000
3,000
4,000
5,000
Unique ionizable lipids